

Evaluation of Polyomavirus BK infection in oncogenic patients

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Summary

Polyomavirus BK, a double-stranded DNA virus, is a member of the Polyomaviridae family that is known to infect humans. Transmission of Polyomavirus BK occurs during childhood via oral or respiratory tracts. Over 80% of adult population is seropositive, but the virus remains latent in the renourinary tract, brain, liver, lymphoid tissues. Certain conditions of the host, especially immunosuppression, can allow Polyomavirus BK to reactivate from the persistent subclinical state to an active lytic infection, resulting in serious disease such as tumors of cells that do not support productive infections. The goal of this cross-sectional study was to evaluate BK Polyomavirus in the cerebrospinal fluid of oncogenic hospitalized patients. 120 cerebrospinal fluid specimens were screened for the BK Polyomavirus DNA. Polymerase chain reaction with specific primers was carried out. PCR products were electrophoresed in 2% gel agarose and the appearance of a 173 base band determined BKV positivity. Positive PCR for BK virus was seen in four patients (3.33%). This finding is very important. Based on the findings, causes of cancer in

the patients can be due to the presence of the virus, or the reactivation of the virus after immunosuppression. Therefore, further studies are needed to investigate the population studied and to evaluate the virus in urine samples and blood samples.



Key words

Polyomavirus BK, Oncogenic patients, Cerebrospinal fluid

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Introduction

Polyoma virus BK is a member of the family Polyomaviridae, a group of small non-enveloped viruses characterized by similar genome structures.¹ The virus was first discovered in 1971 in the urine of a kidney transplant patient with ureteral stenosis.² From the start of their recognized existence, the human Polyomaviruses were linked to important disease.³ It is assumed that the virus establishes a harmless life-long latent infection in the host. It has been associated with upper respiratory disease in children either by the oral or respiratory route and 50% of children are seropositive by the age of 3 and 91% are seropositive by the age of 9.⁴ Through primary infection,^{5,6} different host cells,⁷ including brain, liver, lymphoid tissues and renourinary tract, are exposed to the polyoma BK viruses.⁸ Such a reactivation is characterized by PCR (DNA) detection of the virus in severely immunocompromised hosts.⁹ Thus, Polyomavirus BK is an opportunistic pathogen in immunocompromised patients.¹⁰ Major patient groups such as allograft transplant recipients and oncogenic patients are highly susceptible to BK virus reactivation.¹¹ Recent data reveal that, the BK virus is the essential cause of renal losses in kidney transplanted patients.¹² Nephritis, cancers and urethral stenosis are the most important polyoma BK virus complications.^{13,14} The aim of this study was polyoma BK virus detection in cerebrospinal fluid specimens of oncogenic patients.

Materials and Methods

Sampling

In this cross-sectional study, 120 oncogenic patients in Kerman hospital, Kerman, Iran were enrolled. Cerebrospinal fluid samples were obtained under sterile conditions by lumbar puncture. All specimens were kept at -20°C until they were tested. Vials of 1 to 2 ml of each specimen were shipped to our laboratory in sterile tubes for performance of polymerase chain reaction (PCR) and processed immediately.

Polyomavirus BK DNA Detection

Polyomavirus BK genomic DNA was extracted from the cerebrospinal fluids using QIAamp DNA Mini Kit (Catalog No. DN8115C). The PCR were carried out in a DNA thermal cycler (Analytik Jena, Germany) in 50 μl of a reaction mixture. The reaction mixture was prepared in a separate clean area using a set of dedicated pipettes. For one 50 μl polymerase chain reaction, the mixture PCR master mix (Cinaclone, Iran) (containing H_2O ultrapure 34.60 μl , 10 $\times$ PCR Buffer 2.50 μl , dNTP (10 Mm) 1.00 μl , forward primer (20 pmole/ μl) 1.00 μl , reverse primer (20 pmole/ μl) 1.00 μl , taq (5 U/ μl) 0.40 μl , and MgCl_2 (50 Mm) 3.00 μl . A 46 μl volume of the reaction mixture was dispensed into each microtube. The microtube were then taken to another clean area where the appropriate DNA samples (4 μl) were added to each tube. Negative and positive controls were used in each run. The thermal protocol consisted

35 cycles of initial denaturation at 94°C for 5 min, denaturing at 94°C for 60 s, annealing at 55°C for 30 s, and extension at 72°C for 1 min, followed by final extension 72°C for 10 min and final storage at 4°C.¹⁵⁻¹⁸ A negative control did not have template DNA and contained of polymerase chain reaction master mix, sets of primers and nuclease free water. The primers used for the amplifications named BK LT antigen are presented in table 1. The primers amplified a 176 bp of the gene encoding the LT antigen. The amplicons were analyzed by electrophoresis in 2.0% agarose gel, containing safe stain dye (Gene Tek, Cat. No. GT-SD1000) for 30-40 min at 90v, and examined under ultraviolet light using a UV light transilluminator (UV Star, Biometra, Germany). The safe stain dye is a safe alternative to Ethidium bromide (EB) stain used for de-

tection of nucleic acids in the agarose gels.

Results

Polyomavirus BK DNA was detected in the cerebrospinal fluid of four patients. Agarose gel electrophoresis of the PCR products revealed bands of the appropriate size, according to sequence data for BKV (176bp). The polymerase chain reaction product of the positive samples is re-electrophoresed and the result have been shown in Figure 1.

Discussion

Diseases caused by polyomaviruses are characteristically observed in immunosuppressed people, who are not able to mount a protective immune response

Table 1 The sequences BK LT antigen primers

Target gene	Primer sequences	Amplicon size(bp)
BK LT antigen	5'-AGTCTTTAGGGTCTTCTACC-3'	176
	5'-GGTGCCAACCTATGGAACAG-3'	

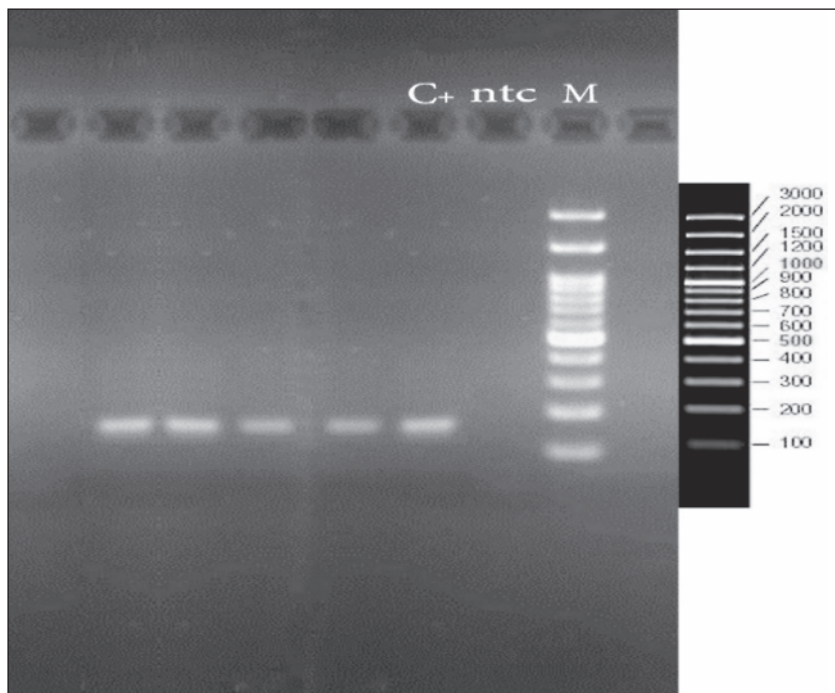


Figure 1 Detection of PCR-amplified BKV DNA in agarose gels. Lane M contains size markers (100-base-pair ladder). Amplified BKV resulted in sequences of 176 bp, NTC: Negative control, C+: Positive control.

to control the viral replication.¹⁹ Polyomavirus BK DNA can be identified in bone marrow and in peripheral blood lymphocytes of the B cell lineage.²⁰ It is possible that the introduction of such infectious lymphoid cells into the central nervous system may result in viral persistence in the area. Reactivation of Polyomavirus BK could occur due to immunosuppression following cancer and the use of immunosuppressive drugs.^{10,21} Therefore, BKV in these patients may be due to primary or reactivated infection.¹⁴ In the present research, positive polymerase chain reaction for Polyomavirus BK was seen in four patients (3.33%). This finding is very important. Observation of the virus BK in the cerebrospinal fluid sample of patients can be a sign of reactivation of the virus in the body, or since the virus belongs to the oncogene viruses group, it can be considered a cause of cancer in these patients. Behzad-Behbahani and his colleagues detected BK virus DNA in cerebrospinal fluid samples of three immunocompetent patients presenting mild encephalitis and in two immunocompromised patients without neurolo-

gical symptoms.²² Although polyoma BK virus infections of the CNS are rare, there is now sufficient evidence for its occurrence in immunocompromised patients and the virus should be considered in such patients with CNS and kidney disorders.²³ In conclusion, Polyoma virus BK was recently found to be an opportunistic central nervous system infectious agent in the immunocompromised host. Detection of polyomaviruses BK virus DNA in the cerebrospinal fluid and brain tissue of patients suggest that this virus has also some neurotropism. This virus must be taken into consideration in the differential diagnosis of central nervous system disorders in cancer, AIDS and transplant patients.

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Conflict of interest

The authors have no potential conflicts of interest.



Περίληψη

Αξιολόγηση της λοίμωξης από τον πολυόμα ιό BK Polyomavirus BK σε ογκολογικούς ασθενείς

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Ο πολυόμα ιός BK, με διπλή έλικα DNA, αποτελεί μέλος της οικογένειας των ιών πολυόμα που είναι γνωστό ότι προκαλούν λοίμωξη στον άνθρωπο. Η μετάδοση των polyoma (πολυόμα) ιών BK συμβαίνει κατά την διάρκεια της παιδικής ηλικίας μέσω της στοματικής ή αναπνευστικής οδού. Άνω των 80% του ενήλικου πληθυσμού είναι οροθετικό, ενώ ο ιός παραμένει σε λανθάνουσα κατάσταση στο νεφρο-ουρητηρικό σύστημα, τον εγκέφαλο, το ήπαρ και τον λεμφικό ιστό.

Σε ορισμένες καταστάσεις του ξενιστού ειδικά σε ανοσοκατεσταλμένους, είναι δυνατό ο polyomavirus BK να επαναδραστηριοποιηθεί από την εμμένουσα υποκλινική κατάσταση σε ενεργό καταστροφική λοίμωξη, με αποτέλεσμα σοβαρή νόσο όπως όγκοι (κακοήθειες) από κύτταρα που δεν προάγουν παραγωγικές λοιμώξεις. Ο στόχος αυτής της συγχρονικής μελέτης ήταν να αξιολογήσει των BK polyomavirus στο εγκεφαλονωτιαίο υγρό νοσηλευόμενων στο νοσοκομείο ογκολογικών ασθενών.

Σε 120 δείγματα εγκεφαλονωτιαίου υγρού αναζητήθηκε το DNA των BK polyomavirus. Ο έλεγχος έγινε με την αντίδραση της αλυσιδωτής πολυμεράσης με ειδικούς εκκινητές. Τα προϊόντα της PCR ηλεκτροφορήθηκαν σε 2% γέλης αγαρόζης και η εμφάνιση μιας ταινίας 173 βάσεων καθόριζε την θετικότητα για BKV. Θετική PCR για BK ιό διαπιστώθηκε σε τέσσερις ασθενείς (3,33%).

Με βάση τα ευρήματα της μελέτης η αιτία του καρκίνου στους ασθενείς μπορεί να οφείλεται στην παρουσία του ιού ή την επανεργοποίηση του ιού λόγω της ανοσοκαταστολής. Περαιτέρω μελέτες απαιτούνται για την διερεύνηση των ασθενών της μελέτης και για την αξιολόγηση του ιού σε δείγματα ούρων και αίματος.



Λέξεις κλειδιά

Πολυόμα-BK (polyomavirus), ογκολογικού ασθενείς, εγκεφαλονωτιαίο υγρό



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